Deformations of associative algebras

The Hochschild complex. Let us consider a deformation of an associative algebra $(\mathcal{A}, \cdot)$ of the form

$$a \star b = a \cdot b + \hbar\, \phi(a,b) + \mathcal{O}(\hbar^2)\,, \qquad a, b \in \mathcal{A}\,, \tag{1}$$

where $\hbar$ is a formal parameter. The condition that this defines a new associative product on $\mathcal{A}$ reads

$$(a \star b) \star c - a \star (b \star c) = -\hbar \big(a \cdot \phi(b,c) - \phi(a \cdot b, c) + \phi(a, b \cdot c) - \phi(a,b) \cdot c\big) + \mathcal{O}(\hbar^2)\,, \tag{2}$$

for any $a, b, c \in \mathcal{A}$. For it to be satisfied at first order in $\hbar$, we therefore find that ϕ should satisfy a constraint involving various composition of it with the undeformed product $\cdot$ of $\mathcal{A}$. For this first order, or infinitesimal, deformation to be non-trivial, it should not be equivalent to the original product. Recall that the product $\star$ is equivalent to $\cdot$ if there exists an operator $T : \mathcal{A}[\![\hbar]\!] \to \mathcal{A}[\![\hbar]\!]$ of the form

$$T = \mathbf{1} + \hbar\, \tau + \mathcal{O}(\hbar^2)\,, \tag{3}$$

which satisfies

$$T(a \star b) - T(a) \cdot T(b) = \hbar \big(\phi(a,b) - a \cdot \tau(b) + \tau(a \cdot b) - \tau(a) \cdot b\big) + \mathcal{O}(\hbar^2)\,, \tag{4}$$

for any $a, b \in \mathcal{A}$. This gives us another condition for ϕ, which is that it should not be given by a particular combination of a map $\tau : \mathcal{A} \to \mathcal{A}$ composed in various way with the undeformed product. To summarise, ϕ defines a non-trivial infinitesimal deformation of $\cdot$ if and only if

(i) It verifies

$$a \cdot \phi(b,c) - \phi(a \cdot b, c) + \phi(a, b \cdot c) - \phi(a,b) \cdot c = 0\,, \tag{5}$$

for any $a, b, c \in \mathcal{A}$;

(ii) It is not obtained from

$$\phi(a,b) \neq a \cdot \tau(b) + \tau(a \cdot b) - \tau(a) \cdot b\,, \tag{6}$$

for any $a, b \in \mathcal{A}$.

These two conditions are sensibly different from one another: the first one is an identity, a property, that ϕ should obey, whereas the second one is a requirement on the form of ϕ, which excludes trivial solution to the first condition. This is reminiscent of a cohomological problem, wherein one tries to find cocycle, i.e. cochains which are annihilated by the differential of a complex, so that they verify a given equation, and one excludes coboundary, i.e. cochains which trivially satisfy the cocyle condition, by virtue of the fact that they are in the image of the differential, and hence take a particular form. We can try to make this analogy more precise by introducing the maps

$$\mathrm{Hom}_{\mathbb{R}}(\mathcal{A}, \mathcal{A}) \xrightarrow{\delta^0} \mathrm{Hom}_{\mathbb{R}}(\mathcal{A} \otimes \mathcal{A}, \mathcal{A}) \xrightarrow{\delta^1} \mathrm{Hom}_{\mathbb{R}}(\mathcal{A} \otimes \mathcal{A} \otimes \mathcal{A}, \mathcal{A})\,, \tag{7}$$

where

$$(\delta^0 \tau)(a,b) = a \cdot \tau(b) + \tau(a \cdot b) - \tau(a) \cdot b\,, \tag{8}$$

and

$$(\delta^1 \phi)(a,b,c) = a \cdot \phi(b,c) - \phi(a \cdot b, c) + \phi(a, b \cdot c) - \phi(a,b) \cdot c\,, \tag{9}$$

and indeed check that the composition vanishes,

$$\delta^1 \circ \delta^0 = 0\,, \tag{10}$$

identically, simply by virtue of the fact that $(\mathcal{A}, \cdot)$ is associative. This leads us to the following definition of the Hochschild complex associated to an associative algebra [1].

Definition 1 [Hochschild complex]**.** Let $(\mathcal{A},\cdot)$ be an associative algebra, and $(M,\triangleright,\triangleleft)$ a bimodule over it. The associated Hochschild complex is the cochain complex

$$\big(\boldsymbol{C}^\bullet(\mathcal{A},M),\delta_{\text{H}}\big)\,,\qquad \boldsymbol{C}^n(\mathcal{A},M):=\text{Hom}_{\mathbb{R}}(\mathcal{A}^{\otimes(n+1)},M)\,,\qquad n\geq -1\,, \tag{11}$$

where the differential δ_{H} is defined as

$$\begin{aligned}(\delta_{\text{H}}\phi)(a_0,\dots,a_n):=a_0\triangleright\phi(a_1,\dots,a_n)+\sum_{k=1}^{n}(-1)^k\,\phi(a_0,\dots,a_{k-1}\cdot a_k,\dots,a_n)\\ -(-1)^n\,\phi(a_0,\dots,a_{n-1})\triangleleft a_n\,,\end{aligned} \tag{12}$$

for any $a_0,\dots,a_n\in\mathcal{A}$.

Let have a look at the first cohomology groups.

- By convention, $\mathcal{A}^{\otimes 0}=\mathbb{R}$ to that elements of degree -1 are linear maps $\mathbb{R}\to M$, which simply single out one element $m\in M$ of the bimodule.[1] On top of that, since it is the lowest degree of the complex, the cohomology is simply constituted of cocycles,

$$H^{-1}(\delta_{\text{H}})=\big\{m\in M\mid a\triangleright m-m\triangleleft a=0\,,\ \forall a\in\mathcal{A}\big\}\,, \tag{13}$$

  which correspond to elements of M on which the left and right actions of $\mathcal{A}$ agree. In other words, the cohomology in degree -1 is the center of the bimodule M, usually denoted by $H^{-1}(\delta_{\text{H}})\cong\mathcal{Z}_{\mathcal{A}}(M)$.

- In degree 0, the Hochschild cohomology is given by

$$\begin{aligned}H^0(\delta_{\text{H}})=\big\{\tau:\mathcal{A}\to M\mid\tau(a\cdot b)=\tau(a)\triangleleft b+a\triangleright\tau(b)\,,\ \forall a,b\in\mathcal{A}\,,\\ \text{and }\ \tau(a)\neq m\triangleleft a-a\triangleright m\,,\ \text{with }\ m\in M\big\}\end{aligned} \tag{14}$$

  i.e. maps from $\mathcal{A}$ to M, which are derivations, and not of the form of a commutator of the left and right actions of $\mathcal{A}$ on a given element of M. Such derivations are called *inner*, and hence cohomology in degree 0 consists of the *outer* derivations of $\mathcal{A}$ (valued in M), $H^0(\delta_{\text{H}})\cong\text{Der}_{\text{out}}(\mathcal{A},M)$.

- Finally, when the bimodule $M=\mathcal{A}$ is the algebra itself, the cohomology in degree 1 is

$$\begin{aligned}H^1(\delta_{\text{H}})=\big\{\phi:\mathcal{A}\otimes\mathcal{A}\to\mathcal{A}\mid a\cdot\phi(b,c)-\phi(a\cdot b,c)+\phi(a,b\cdot c)-\phi(a,b)\cdot c=0\,,\\ \forall a,b,c\in\mathcal{A}\ \text{and}\ \ \phi(a,b)\neq a\cdot\tau(b)-\tau(a\cdot b)+\tau(a)\cdot b\,,\ \text{with}\ \ \tau:\mathcal{A}\to\mathcal{A}\big\}\end{aligned} \tag{15}$$

  i.e. it consists in equivalence classes of infinitesimal deformations of $(\mathcal{A},\cdot)$, as we previously remarked.

Lemma 1. *Let $(\mathcal{A},\cdot)$ be an associative algebra and $(M,\triangleleft,\triangleright)$ a* symmetric *bimodule over it (i.e. a bimodule for which the left and right action agree, $a\triangleleft m=m\triangleright a$ for any $a\in\mathcal{A}$ and $m\in M$). Then all multiderivations of $\mathcal{A}$ valued in the bimodule M are Hochschild cocycles.*

Proof. Let us define

$$L^k:\boldsymbol{C}^{n-1}(\mathcal{A},M)\longrightarrow\boldsymbol{C}^n(\mathcal{A},M)\,,\qquad 0\leq k\leq n-1\,, \tag{16}$$

via

$$(L^k\Phi)(-,a,b,-)=a\triangleright\Phi(\underset{\uparrow}{-},b,-)+\Phi(\underset{\uparrow}{-},a,-)\triangleleft b-\Phi(\underset{\uparrow}{-},a\cdot b,-)\,,\qquad a,b\in\mathcal{A}\,, \tag{17}$$
$$\qquad\qquad\ k\qquad\qquad\quad k\qquad\qquad\quad k$$

where the arguments are inserted in the $(k+1)$th entry of the cochain Φ. This map is the Leibniz rule for the $(k+1)$th argument of Φ, i.e. it vanishes if and only if Φ is a M-valued derivation in its $(k+1)$th entry. A direct computation yields

$$\begin{aligned}\sum_{k=0}^{n-1}(-1)^k\,(L^k\Phi)(a_0,\dots,a_n)=(\delta_{\text{H}}\Phi)(a_0,\dots,a_n)\\ -\sum_{k=1}^{n-1}(-1)^k\,\big(a_k\triangleright\Phi(-,\hat{a}_k,-)-\Phi(-,\hat{a}_k,-)\triangleleft a_k\big)\,,\end{aligned} \tag{18}$$

[1] Indeed, if $f:\mathbb{R}\to M$ is a linear map, then for any $\lambda\in\mathbb{R}$, one has $f(\lambda)=f(\lambda\times 1)=\lambda\,f(1)$, so that f is completely determined by its value on 1, which defines an element $m:=f(1)\in M$ of the bimodule.

where the hat signifies that the argument is omitted. If Φ is a multiderivation, then $L^k\Phi$ vanishes for all $0 \leq k \leq n-1$, and if on top of that Φ takes values in a *symmetric* bimoudle M, the second term on the right hand side of the above equation also vanishes, hence Φ is cocycle. ■

The Hochschild complex of a commutative algebra. Now let us articulate some refined results that can be obtained in the special case where the algebra $(\mathcal{A}, \cdot)$ is not only associative but also commutative.

Proposition 2 (Semi-classicalisation)**.** *The antisymmetric part of equivalence classes of infinitesimal deformations of a commutative and associative algebra $(\mathcal{A}, \cdot)$ are in 1-to-1 correspondence with Poisson algebra structures on $\mathcal{A}$.*

Proof. Let $\star = \cdot + \hbar\,\phi + \mathcal{O}(\hbar^2)$ be a deformation of the commutative and associative product $\cdot$ of $\mathcal{A}$, and define the $\star$-commutator

$$[a, b]_\star := a \star b - b \star a\,, \tag{19}$$

for $a, b \in \mathcal{A}$. As a commutator defined from an associative product, it satisfies the Leibniz rule

$$[a, b \star c]_\star = [a, b]_\star \star c + b \star [a, c]_\star\,, \qquad a, b, c \in \mathcal{A}\,, \tag{20}$$

and the Jacobi identity

$$[a, [b, c]_\star]_\star = [[a, b]_\star, c]_\star + [b, [a, c]_\star]_\star\,, \qquad a, b, c \in \mathcal{A}\,, \tag{21}$$

which, once expanded at first and second order in $\hbar$ respectively, imply that the antisymmetric part of ϕ, i.e.

$$\{a, b\}_\phi := \tfrac{1}{2}\left(\phi(a, b) - \phi(b, a)\right), \qquad a, b \in \mathcal{A}\,, \tag{22}$$

satisfies both the Leibniz rule (with respect to $\cdot$) and the Jacobi identity, respectively. In other words, $(\mathcal{A}, \cdot, \{-, -\})$ is a Poisson algebra.

Suppose now that $\star$ and $\star'$ are two equivalent deformations of $\cdot$ related by $T : \mathcal{A}[\![\hbar]\!] \to \mathcal{A}[\![\hbar]\!]$ and whose first orders in $\hbar$ are respectively given by ϕ and ϕ'. Expanding the condition

$$T([a, b]_{\star'}) = [T(a), T(b)]_\star\,, \qquad a, b \in \mathcal{A}\,, \tag{23}$$

at first order in $\hbar$ leads to

$$\{a, b\}_{\phi'} = \{a, b\}_\phi\,, \qquad a, b \in \mathcal{A}\,, \tag{24}$$

i.e. two equivalent deformations of $\cdot$ induce the same Poisson structure on $\mathcal{A}$. ■

For the case of interest for us which $\mathcal{A} = \mathcal{C}^\infty(M)$, the algebra of functions on a smooth manifold, we learn that star-products are infinitesimally controlled by Poisson structures. The previous proposition also tells us that, when $(\mathcal{A}, \cdot)$ is commutative, the antisymmetric part of an infinitesimal deformation is always a bi-derivation (i.e. a derivation in each one of its arguments). In fact we can show a little more.

Lemma 3. *Let $(\mathcal{A}, \cdot)$ be a commutative and associative algebra. The following statements hold true in the adjoint Hochschild complex (i.e. when $M = \mathcal{A}$):*

(*i*) *All multiderivations of $\mathcal{A}$ are cocycles;*

(*ii*) *The antisymmetric part of a coboundary vanishes;*

(*iii*) *An antisymmetric cochain is a multiderivation if and only if it is a cocycle.*

Proof.

(*i*) This follows immediately from the previous Lemma 1 and the fact that the center of a commutative algebra vanishes.

(*ii*) This follows from a direct computation: antisymmetrising the formula for the action of the Hochschild differential over all arguments produces linear combinations of terms wherein only the commutator of elements of $\mathcal{A}$ appear, which all vanish by virtue of the fact that $\mathcal{A}$ is commutative.

(*iii*) The implication from multiderivations to cocycles has been proved in (*i*), so we can focus on proving the converse. For an antisymmetric cochain, the Leibniz rule for the $(k+1)$th argument defined in (17) is related to that of the first argument,

$$L^k(a_0,\dots,a_n) = (-1)^k L^0(a_k, a_{k+1}, a_0, \dots, a_{k-1}, a_{k+2}, \dots, a_n)\,, \tag{25}$$

and since a commutative algebra $\mathcal{A}$ is a symmetric bimodule over itself, the relation (18) reads

$$(\delta_{\text{H}}\phi)(a_0,\dots,a_n) = \sum_{k=0}^{n-1} L^0(a_k, a_{k+1}, a_0, \dots, a_{k-1}, a_{k+2}, \dots, a_n)\,. \tag{26}$$

It can be 'inverted' in the sense that one can express the Leibniz rule as a linear combination of cocycle conditions,

$$L^0(a_0,\dots,a_n) = \tfrac{1}{n}\sum_{k=0}^{n}(-1)^{kn}(1-n\delta_{k,1})(\delta_{\text{H}}\phi)(a_k,\dots,a_n,a_0,\dots,a_{k-1})\,, \tag{27}$$

which therefore proves the claim.

■

Gerstenhaber product. Now let us go back to the general case, i.e. we no longer assume $(\mathcal{A},\cdot)$ to be commutative. There is one more structure that can be defined on the Hochschild complex associated $\mathcal{A}$, called the Gerstenhaber product [2].

Definition 2 [**Gerstenhaber product**]**.** Let $\mathcal{A}$ be a vector space, and

$$\boldsymbol{C}^n(\mathcal{A},\mathcal{A}) := \text{Hom}_{\mathbb{R}}(\mathcal{A}^{\otimes(n+1)},\mathcal{A})\,, \qquad n \geq -1\,, \tag{28}$$

as in Definition 1. The operation

$$\bullet_{\text{G}} : \boldsymbol{C}^m(\mathcal{A},\mathcal{A}) \otimes \boldsymbol{C}^n(\mathcal{A},\mathcal{A}) \longrightarrow \boldsymbol{C}^{n+m}(\mathcal{A},\mathcal{A})\,, \tag{29}$$

defined by

$$(\phi \bullet_{\text{G}} \psi)(a_0,\dots,a_{n+m}) := \sum_{k=0}^{m}(-1)^{kn}\,\phi(a_0,\dots,a_{k-1},\psi(a_k,\dots,a_{k+n}),a_{k+n+1},\dots,a_{m+n})\,, \tag{30}$$

for any $a_0,\dots,a_{n+m}$, is called the Gerstenhaber product.

Schematically, the Gerstenhaber product takes the form

$$\phi \bullet_{\text{G}} \psi = \sum_{k=0}^{|\phi|}(-1)^{k|\psi|}\,\phi(\underset{\overset{\uparrow}{k}}{-},\psi(-),-)\,, \tag{31}$$

where we replaced the number of arguments of ϕ and ψ with their degrees, denoted respectively $|\phi|$ and $|\psi|$, in the Hochschild complex. In plain words, it is a particular linear combination of the insertions of ψ in the different entries of ϕ. The sign in front of each term can be justified as follows: recall that an element of $\mathcal{A}$ can equivalently be thought of as a linear map $\mathbb{R}\to\mathcal{A}$, i.e. an element of $\boldsymbol{C}^{-1}(\mathcal{A},\mathcal{A}) = \text{Hom}_{\mathbb{R}}(\mathbb{R},\mathcal{A}) \cong \mathcal{A}$. As a consequence, elements of $\mathcal{A}$ can be seen as elements of degree -1 in the Hochschild complex. The sign $(-1)^{k|\psi|}$ before the insertion of ψ in the $(k+1)$th entry of ϕ can then be justified by the fact that ψ had to 'jump' over k elements of degree -1. This is consistent with the sign rule

$$(f\otimes g)(x\otimes y) = (-1)^{|g||x|}\,f(x)\otimes g(y)\,, \tag{32}$$

for evaluating the tensor product of any two maps of graded vector spaces $f : V_1 \to W_1$ and $g : V_2 \to W_2$ on a pair of elements $x \in V_1$ and $y \in V_2$ (see e.g. this note for more details on sign rules).

This operation however *not* associative, but it verifies a slightly weaker property, which makes it a pre-Lie structure.

Definition 3 [Pre-Lie structure]**.** Let V be a $\mathbb{Z}$-graded vector space, endowed with a bilinear operation $\bullet : V \otimes V \to V$. If its associator $A : V \otimes V \otimes V \to V$, defined as

$$A(x, y, z) := (x \bullet y) \bullet z - x \bullet (y \bullet z)\,, \qquad x, y, z \in V\,, \tag{33}$$

is graded symmetric in its last two arguments, meaning $A(x, y, z) = (-1)^{|y||z|}\, A(x, z, y)$, then $(V, \bullet)$ is called a ($\mathbb{Z}$-graded) pre-Lie algebra.

(∗) Remark 1. In fact, this above definition is that of a right pre-Lie structure, a left pre-Lie structure consisting of a bilinear operation whose associator is (graded) symmetric in its first two arguments.

The fact that the above structure is called pre-Lie is justified by the following lemma.

Lemma 4. *If $(V, \bullet)$ is a (graded) pre-Lie algebra, then the (graded) commutator defines a (graded) Lie bracket on V.*

Proof. A direct computation shows that the Jacobi identity for the graded commutator

$$J(x, y, z) := [x, [y, z]_\bullet]_\bullet - [[x, y]_\bullet, z]_\bullet - (-1)^{|x||y|}\, [y, [x, z]_\bullet]_\bullet\,, \qquad x, y, z \in V\,, \tag{34}$$

can be written in terms of the associator of $\bullet$, as

$$\begin{aligned} J(x, y, z) = -A(x, y, z) + (-1)^{|y||z|}\, A(x, z, y) + (-1)^{|x||y|}\, \big(A(y, x, z) - (-1)^{|x||z|}\, A(y, z, x)\big) \\ - (-1)^{|z|(|x|+|y|)}\, \big(A(z, x, y) - (-1)^{|x||y|}\, A(z, y, x)\big)\,, \end{aligned} \tag{35}$$

which vanishes whenever $(V, \bullet)$ is a (right, $\mathbb{Z}$-graded) pre-Lie algebra, hence the above lemma. ∎

The defining property of a pre-Lie structure, i.e. the fact that the associator is graded-symmetric in its last two arguments, is the aforementioned property, 'weaker than associativity', verified by the Gerstenhaber product.

Lemma 5. *The Gerstenhaber product is a (right) pre-Lie structure on $\boldsymbol{C}^\bullet(\mathcal{A}, \mathcal{A})$.*

Proof. This can be shown by direct computation, we will just sketch the mechanism. Given three cochains $\phi, \varphi, \psi \in \boldsymbol{C}^\bullet(\mathcal{A}, \mathcal{A})$, their Gerstenhaber product schematically reads

$$(\phi \bullet_{\mathsf{G}} \varphi) \bullet_{\mathsf{G}} \psi \sim \sum \phi\Big(-, \varphi\big(-, \psi(-), -\big), -\Big) \tag{36}$$

$$+ \phi\big(-, \varphi(-), -, \psi(-), -\big) + \phi\big(-, \psi(-), -, \varphi(-), -\big)\,, \tag{37}$$

where the sum runs over all possible insertions of the cochains in one another, as denoted. The first line of the right hand side of the above equation corresponds to $\phi \bullet_{\mathsf{G}} (\varphi \bullet_{\mathsf{G}} \psi)$, so that the associator of the Gerstenhaber product reduces to terms of the form displayed in the second line. The latter are symmetric under the exchange of φ and ψ, as required of a right pre-Lie structure. ∎

As a consequence of Lemma 4 that the Hochschild complex is also a graded Lie algebra.

Corollary 6. *The Gerstenhaber bracket, defined as the graded commutator of the Gerstenhaber product,*

$$[\phi, \psi]_{\mathsf{G}} := \phi \bullet_{\mathsf{G}} \psi - (-1)^{|\phi||\psi|}\, \psi \bullet_{\mathsf{G}} \phi\,, \qquad \phi, \psi \in \boldsymbol{C}^\bullet(\mathcal{A}, \mathcal{A})\,, \tag{38}$$

defines a Lie bracket on the Hochschild complex.

The Hochschild differential δ_{H} can be re-written in terms of the Gerstenhaber bracket. To see this, first notice that the undeformed product can be seen as an element of the Hochschild complex of degree 1, that we will denote $m \in \boldsymbol{C}^1(\mathcal{A}, \mathcal{A}) = \mathrm{Hom}_{\mathbb{R}}(\mathcal{A} \otimes \mathcal{A}, \mathcal{A})$. Taking its Gerstenhaber product with a cochain $\phi \in \boldsymbol{C}^{n-1}(\mathcal{A}, \mathcal{A})$ from the left and from the right respectively reads

$$(\phi \bullet_{\mathsf{G}} m)(a_0, \dots, a_n) = \sum_{k=0}^{n-1} (-1)^k\, \phi(a_0, \dots, a_{k-1} \cdot a_k, \dots, a_n)\,, \tag{39}$$

and

$$(m \bullet_{\mathsf{G}} \phi)(a_0, \dots, a_n) = \phi(a_0, \dots, a_{n-1}) \cdot a_n - (-1)^n\, a_0 \cdot \phi(a_1, \dots, a_n)\,, \tag{40}$$

so that

$$\delta_{\mathsf{H}}\phi = -[\phi, m]_{\mathsf{G}} = (-1)^{|\phi|}\,[m, \phi]_{\mathsf{G}}\,, \tag{41}$$

i.e. the Hochschild differential is inner in the graded Lie algebra that is $\big(\boldsymbol{C}^\bullet(\mathcal{A},\mathcal{A}), [-,-]_{\mathsf{G}}\big)$. In particular, this implies that δ_{H} differentiates the Gerstenhaber bracket, in the sense that it verifies

$$\delta_{\mathsf{H}}[\phi,\psi]_{\mathsf{G}} = (-1)^{|\psi|}\,[\delta_{\mathsf{H}}\phi,\psi]_{\mathsf{G}} + [\phi,\delta_{\mathsf{H}}\psi]_{\mathsf{G}}\,, \tag{42}$$

as a consequence of the Jacobi identity. Notice however that the signs appearing in this equation are not natural: one would expect that these signs to be recovered by the aforementioned Koszul sign convention, i.e. to reflect the fact that δ_{H} is a degree 1 derivation of the Gerstenhaber bracket. This is simply a consequence of the fact that the Hochschild differential acts as the adjoint action by m only up to a sign. We will therefore redefine the Hochschild differential as

$$\delta_{\mathsf{H}} := [m, -]_{\mathsf{G}}\,, \tag{43}$$

from now on. The fact that δ_{H} is a differential can also be recast in terms of the Gerstenhaber bracket,

$$\delta_{\mathsf{H}}^2 = 0 \qquad \Longleftrightarrow \qquad \tfrac{1}{2}\,[[m,m]_{\mathsf{G}}, -]_{\mathsf{G}} = 0\,, \tag{44}$$

upon using the Jacobi identity. Evaluating the Gerstenhaber bracket of m with itself,

$$\tfrac{1}{2}\,[m,m]_{\mathsf{G}}(a,b,c) = (a\cdot b)\cdot c - a\cdot(b\cdot c) = 0\,, \qquad a,b,c\in\mathcal{A}\,, \tag{45}$$

one finds that it vanishes as a consequence of the associativity of m. In other words, we recover in a fairly concise form the fact that δ_{H} squares to zero as a consequence of m being associative.

The previous computation also reveals an important lesson: associative products on a vector space $\mathcal{A}$ are in bijection with elements $\star\in\boldsymbol{C}^1(\mathcal{A},\mathcal{A})$, i.e. linear maps from $\mathcal{A}\otimes\mathcal{A}$ to $\mathcal{A}$, which verify

$$\tfrac{1}{2}\,[\star,\star]_{\mathsf{G}} = 0\,, \tag{46}$$

encoding the associativity condition. In the case of a star product, i.e. a formal deformation of another associative product m on $\mathcal{A}$, such an element of $\boldsymbol{C}^1(\mathcal{A},\mathcal{A})[\![\hbar]\!]$ should be written as

$$\star = m + \Phi\,, \qquad \Phi\in\hbar\boldsymbol{C}^1(\mathcal{A},\mathcal{A})[\![\hbar]\!]\,, \tag{47}$$

and the associativity condition reads

$$0 = \tfrac{1}{2}\,[\star,\star]_{\mathsf{G}} = \delta_{\mathsf{H}}\Phi + \tfrac{1}{2}\,[\Phi,\Phi]_{\mathsf{G}}\,, \tag{48}$$

upon using the fact that $\delta_{\mathsf{H}}\equiv[m,-]_{\mathsf{G}}$. This equation is a first example of the Maurer–Cartan equation in a differential graded Lie algebra, both of which will be the subject of the next paragraphs.

References

[1] G. Hochschild, "On the Cohomology Groups of an Associative Algebra," *Annals of Mathematics* **46** (1945) no. 1, 58–67.

[2] M. Gerstenhaber, "The Cohomology Structure of an Associative Ring," *Annals of Mathematics* **78** (1963) no. 2, 267–288.